SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Deep Learning COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Duration: 1 Hour

QUESTIONS: 5 Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
   H H T H H H T H H T
   Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p)

   Given outcomes:
   H H T H H H T H H T

   Total tosses, n=10
   Number of Heads, x=7
   Number of Tails, 333
   For a sequence of independent coin tosses, the likelihood function is:
   L(p)= p^{x}(1-p)^{n-x}
   Substituting the values:
   L(p)= p^7(1-p)^3
   The MLE of p is obtained by maximizing the likelihood (or log-likelihood), giving the standard result:
   P^=x/n
   Therefore,
   P^=7/10=0.7

   Answer:
   The **Maximum Likelihood Estimate (MLE)** of the probability of getting Heads is
   p^=0.7
   So, based on the observed data, the estimated probability of getting **Heads is 0.7 (70%)**.

2. Out of 50 students, 42 passed an exam.
   Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

   Given:
   Total students, n=50
   Students who passed, x=42
   Assuming each student's result follows a **Bernoulli distribution**, the likelihood function is:
   L(p) = p^x(1-p)^{n-x}
   Substituting the values:

$L(p) = p^{42}(1-p)^8$

The **Maximum Likelihood Estimator (MLE)** for the probability of success (passing) is:

$\hat{P}=x/n$

Substituting the given values:

$\hat{P} = 42/50 \quad =0.84$

Answer:

The **Maximum Likelihood Estimate (MLE)** of the probability that a student passes is:

$\hat{p} \quad =0.84$

Therefore, the estimated probability of a student passing the exam is 0.84 (84%).

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
   Find the MLE for the probability that a bulb is defective.

Given:

- Total bulbs, $n=100$
- Defective bulbs, $x=8$

Assuming each bulb is either **defective** or **non-defective** (Bernoulli distribution), the likelihood function is:

$L(p) = p^x(1-p)^{n-x}$

Substituting the given values:

$L(p) = p^8(1-p)^{92}$

The **Maximum Likelihood Estimator (MLE)** for the probability of a bulb being defective is:

$\hat{p} \quad =x/n$

Substituting the values:

$\hat{p} \quad =8/100 \quad =0.08$

Answer:

The **Maximum Likelihood Estimate (MLE)** of the probability that a bulb is defective is:

$\hat{p} \quad =0.08$

Therefore, the estimated probability of a bulb being defective is 0.08 (8%).

4. A biased die is rolled 60 times. The number 6 appears 18 times.
   Estimate the probability of rolling a 6 using MLE.

Given:

- Total rolls, $n=60$
- Number of times a **6** appears, $x=18$

Treating "rolling a 6" as a **success** (Bernoulli trial), the likelihood function is:

$L(p) = p^x(1-p)^{n-x}$

Substituting the values:

$L(p) = p^{18}(1-p)^{42}$

The **Maximum Likelihood Estimator (MLE)** for the probability of rolling a 6 is:

$\hat{p} = x/n$

Substituting the given values:

$\hat{p} = 18/60 = 0.30$

Answer:

The **Maximum Likelihood Estimate (MLE)** of the probability of rolling a 6 is:

$\hat{p} = 0.30$

Therefore, the estimated probability of rolling a 6 is 0.30 (30%).

5. Given:
   Input (x) = 4
   Weight (w) = 2
   Actual Output = 10
   Prediction:
   $y = w \times x$
   1. Find the predicted output.
      $$y = w \times x = 2 \times 4 = 8$$
   2. Calculate the error.
      $$\text{Error} = \text{Actual Output} - \text{Predicted Output} = 10 - 8 = 2$$
   3. If the gradient is 4 and learning rate is 0.01, find the updated weight.
      $$w_{new} = w - \eta \times \text{gradient} = 2 - (0.01 \times 4) = 2 - 0.04 = 1.96$$

Code of Conduct:

I B.Jahnavi / 192325043 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.Jahnavi

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation